

TOPOLOGICAL STRING THEORY ON CALABI-YAU THREEFOLDS AND THE ROOT DATUM OF THE QUANTUM VERTEX CHIRAL GROUP

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ABSTRACT. We develop the dictionary between topological string theory on a Calabi–Yau threefold X and the quantum vertex chiral group $G(X)$ of Volume III. The A-model free energy $F = \sum_g F_g g_s^{2g-2}$, with F_g encoding Gromov–Witten invariants $N_{g,\beta}$, is identified with the genus expansion of the shadow obstruction tower Θ_{A_X} . The B-model holomorphic anomaly equation of BCOV is reinterpreted as the Maurer–Cartan equation $D\Theta + \frac{1}{2}[\Theta, \Theta] = 0$ projected to genus g . The topological vertex (AKMV) for toric CY3 is shown to factorize as a tensor product of local root data, one per vertex of the toric diagram. The leading genus- g term $F_g = \kappa \cdot \lambda_g$ is the uniform-weight lane of the Vol I shadow obstruction tower; higher terms are shadow corrections from imaginary roots.

CONTENTS

1. INTRODUCTION AND OVERVIEW

Let X be a compact (or appropriately completed) Calabi–Yau threefold. The topological string on X comes in two versions — the A-model and the B-model — related by mirror symmetry $X \leftrightarrow X^\vee$. The central object on both sides is the free energy

$$(1) \quad F(g_s; t) = \sum_{g=0}^{\infty} F_g(t) g_s^{2g-2},$$

where g_s is the string coupling and t denotes moduli (Kähler on the A-side, complex structure on the B-side). The genus- g amplitude F_g is the string-theoretic analogue of the genus- g obstruction $\text{obs}_g(A_X)$ of the quantum vertex chiral group $G(X)$ constructed in Volume III.

The purpose of this note is to make the following identifications precise:

Topological string on X	Quantum vertex chiral group $G(X)$
Free energy F	Shadow obstruction tower Θ_{A_X}
Genus- g amplitude F_g	Genus- g obstruction $\text{obs}_g(A_X)$
GW invariants $N_{g,\beta}$	Shadow amplitudes of $G(X)$ at genus g
BCOV holomorphic anomaly	MC equation $D\Theta + \frac{1}{2}[\Theta, \Theta] = 0$
Topological vertex $C_{\lambda\mu\nu}(q)$	Local root datum at a toric vertex
DT partition function $Z^{\text{DT}}(X)$	Bar-complex Euler product of $G(X)$
Leading F_g on uniform lane	$\kappa(G(X)) \cdot \lambda_g$

The identification $g_s = \hbar$ (string coupling = the quantization parameter of the chiral algebra) and $Q^\beta = e^{-t \cdot \beta}$ (exponentiated Kähler class = root-lattice fugacity) will be implicit throughout.

2. THE A-MODEL TOPOLOGICAL STRING ON CY3

2.1. The A-model partition function. The A-model topological string on a CY3 X localizes onto holomorphic maps $f: \Sigma_g \rightarrow X$ from a genus- g worldsheet to X . The genus- g free energy is a generating function for Gromov–Witten invariants:

$$(2) \quad F_g(t) = \sum_{\beta \in H_2(X, \mathbb{Z})} N_{g,\beta} Q^\beta,$$

where $Q^\beta = e^{-\int_\beta \omega}$ for ω the complexified Kähler form, and $N_{g,\beta}$ is the genus- g degree- β Gromov–Witten invariant:

$$N_{g,\beta} = \int_{[\overline{\mathcal{M}}_{g,0}(X, \beta)]^{\text{vir}}} 1.$$

The virtual fundamental class $[\overline{\mathcal{M}}_{g,0}(X, \beta)]^{\text{vir}}$ has virtual dimension zero for CY3 (since $c_1(X) = 0$), so $N_{g,\beta}$ is a number — a virtual count of genus- g holomorphic curves in class β .

The total free energy

$$F = \sum_{g=0}^{\infty} g_s^{2g-2} \sum_{\beta} N_{g,\beta} Q^\beta$$

is the A-model topological string partition function (at the level of the free energy; the partition function proper is $Z = e^F$).

2.2. Gopakumar–Vafa reformulation. The Gopakumar–Vafa conjecture (now theorem, Ionel–Parker) repackages the GW invariants into integer BPS invariants $n_\beta^g \in \mathbb{Z}$:

$$(3) \quad F = \sum_{g=0}^{\infty} \sum_{\beta \neq 0} \sum_{k=1}^{\infty} \frac{n_\beta^g}{k} \left(2 \sin \frac{k g_s}{2} \right)^{2g-2} Q^{k\beta}.$$

The integers n_β^g count BPS states of charge β and spin g in the M-theory lift. In the language of the quantum vertex chiral group:

- n_β^0 are the genus-0 BPS counts = DT invariants = root multiplicities $\text{mult}(\beta)$ of $G(X)$.

- n_β^g for $g > 0$ are the genus- g refinements of the root multiplicity, living in the g -th shadow correction.

Remark 2.1 (GV invariants as root data). The generating function $\sum_\beta n_\beta^0 Q^\beta$ is the multiplicity-generating function of the imaginary roots of $G(X)$. For the $K3 \times E$ family, these are the Fourier coefficients of $\phi_{0,1}$ (the K3 elliptic genus). For a toric CY3, they are the DT invariants of the quiver with potential (Q_X, W_X) . The higher-genus invariants n_β^g refine the root datum by incorporating information from the Hodge bundle $\mathbb{E}_g \rightarrow \overline{\mathcal{M}}_{g,0}$; in the shadow obstruction tower language, they appear as genus- g shadow corrections to the tree-level root multiplicities.

2.3. GW invariants as shadow amplitudes. The quantum vertex chiral group $G(X) = \Phi(\mathcal{C}_X)$ (where $\mathcal{C}_X$ is the appropriate CY category of X — the Fukaya category on the A-side, or $D^b(\text{Coh}(X^\vee))$ on the B-side via HMS) carries a shadow obstruction tower Θ_{A_X} (Volume I). The genus expansion of Θ_{A_X} is:

$$(4) \quad \Theta_{A_X} = \sum_{g=0}^{\infty} \hbar^{2g-2} \Theta_{A_X}^{(g)},$$

where $\Theta_{A_X}^{(g)} \in C^\bullet(\overline{\mathcal{M}}_{g,n}, B(A_X)^{\otimes n})$ are the genus- g shadow amplitudes — sections of the n -fold tensor product of the bar complex over the moduli space of n -pointed genus- g curves.

Conjecture 2.2 (GW/shadow identification). *For a CY3 X and its quantum vertex chiral group $G(X)$:*

- (i) *The genus- g , n -point shadow amplitude $\Theta_{A_X}^{(g)}|_\beta$ evaluated on the class β recovers the GW descendant invariant:*

$$\langle \Theta_{A_X}^{(g)} \rangle_\beta = \int_{[\overline{\mathcal{M}}_{g,n}(X,\beta)]^{\text{vir}}} \psi_1^{a_1} \cdots \psi_n^{a_n}.$$

- (ii) *The genus- g free energy F_g is the 0-point specialization:*

$$F_g = \langle \Theta_{A_X}^{(g)} \rangle = \text{obs}_g(A_X).$$

- (iii) *The full topological string free energy is the shadow obstruction tower:*

$$F = \Theta_{A_X}$$

under the identification $g_s = \hbar$, $Q^\beta = e^{-t \cdot \beta}$.

Remark 2.3. The identification $F = \Theta_{A_X}$ is of course schematic: F is a formal power series in g_s and Q^β , while Θ_{A_X} is a Maurer–Cartan element in the dgLa governing deformations of the chiral bar complex. The precise statement is that the formal expansion of Θ_{A_X} in the genus grading and the root-lattice grading reproduces F .

3. THE B-MODEL AND THE HOLOMORPHIC ANOMALY EQUATION

3.1. Special geometry of the CY moduli space. Let $X^\vee$ be the mirror CY3 and $\mathcal{M} = \mathcal{M}_{\text{cpx}}(X^\vee)$ its complex structure moduli space, of complex dimension $h = h^{2,1}(X^\vee)$. The B-model free energy F_g is a function (more precisely, a section of a line bundle) on $\mathcal{M}$. The moduli space $\mathcal{M}$ carries the special Kähler structure:

- Special coordinates t^i , $i = 1, \dots, h$;
- The prepotential $\mathcal{F}(t) = F_0(t)$ (the genus-0 free energy);

- The period matrix $\tau_{ij} = \partial_i \partial_j \mathcal{F}$;
- The Weil–Petersson metric $G_{i\bar{j}} = -\partial_i \partial_{\bar{j}} K$ where K is the Kähler potential;
- The Yukawa coupling $C_{ijk} = \partial_i \partial_j \partial_k \mathcal{F}$ (the genus-0 three-point function);
- The propagator S^{ij} satisfying $\partial_{\bar{k}} S^{ij} = \bar{C}_{\bar{k}}^{ij} = e^{2K} G^{i\bar{i}'} G^{j\bar{j}'} \bar{C}_{\bar{k}\bar{i}'\bar{j}'}$.

The propagator S^{ij} is the genus-1 annulus amplitude: it mediates the degeneration of a genus- g surface into lower genera by pinching a cycle.

3.2. The BCOV holomorphic anomaly equation. The central result of Bershadsky–Cecotti–Ooguri–Vafa (BCOV) is that the genus- g free energy, while not holomorphic, satisfies a controlled recursion in g :

$$(5) \quad \partial_i F_g = \frac{1}{2} \bar{C}_i^{jk} \left(D_j D_k F_{g-1} + \sum_{r=1}^{g-1} D_j F_r D_k F_{g-r} \right), \quad g \geq 2.$$

Here $D_i = \nabla_i + (2 - 2g) \partial_i K$ is the covariant derivative incorporating the Kähler connection, and $\bar{C}_i^{jk} = e^{2K} G^{j\bar{j}'} G^{k\bar{k}'} \bar{C}_{i\bar{j}'\bar{k}'}$ is the anti-holomorphic Yukawa coupling raised by the metric. This equation says: the failure of F_g to be holomorphic is controlled by the propagator and the lower-genus data.

Remark 3.1 (The genus-1 anomaly). At genus $g = 1$, the anomaly takes a different form:

$$\partial_i \partial_j F_1 = \frac{1}{2} C_{jkl} \bar{C}_i^{kl} - \left(\frac{\chi(X)}{24} - 1 \right) G_{j\bar{i}},$$

where $\chi(X)$ is the Euler characteristic. In the BCOV formula the coefficient $\chi(X)/24$ controls the genus-1 gravitational anomaly of the *target space* effective theory. This need not equal the modular characteristic $\kappa(A_X)$ of Vol I in general: for $K3 \times E$, $\chi = 0$ while $\kappa = 5$ (see the anomaly cancellation note, AP48). The identification $\kappa = \chi(X)/24$ holds only when the quantum vertex chiral group reduces to its Heisenberg subalgebra.

3.3. BCOV as Maurer–Cartan. The key insight, which connects the B-model to the framework of Volume I, is that the BCOV holomorphic anomaly equation (??) IS the Maurer–Cartan equation

$$(6) \quad D\Theta + \frac{1}{2} [\Theta, \Theta] = 0$$

projected to genus g .

Construction 3.2 (BCOV as MC). The dictionary is as follows.

- (i) **The dgLa.** The relevant dgLa is the BCOV complex

$$\mathcal{L}_X = \text{PV}^{\bullet, \bullet}(X^\vee) = \bigoplus_{p,q} \Omega^{0,q}(X^\vee, \bigwedge^p TX^\vee),$$

the space of polyvector-valued $(0, q)$ -forms on $X^\vee$, with differential $\bar{\partial}$ and bracket given by the Schouten–Nijenhuis bracket wedged with the Dolbeault differential. This is the Kodaira–Spencer dgLa governing deformations of $X^\vee$.

- (ii) **The MC element.** The total free energy

$$\Theta = \sum_{g=0}^{\infty} \hbar^{2g-2} \Theta^{(g)}, \quad \Theta^{(g)} \sim F_g,$$

assembled from all genera, is a MC element of $\mathcal{L}_X[[\hbar]]$.

- (iii) **The genus- g projection.** The MC equation $D\Theta + \frac{1}{2}[\Theta, \Theta] = 0$, expanded in powers of $\hbar^{2g-2}$, gives at genus g :

$$(7) \quad D\Theta^{(g)} + \frac{1}{2} \sum_{r=1}^{g-1} [\Theta^{(r)}, \Theta^{(g-r)}] = 0.$$

This is the BCOV equation (??): the differential D encodes $\bar{\partial}$ plus the Kähler connection, while the bracket $[\Theta^{(r)}, \Theta^{(g-r)}]$ corresponds to the bilinear term $D_j F_r D_k F_{g-r}$ contracted with the propagator $\bar{C}_i^{jk}$.

- (iv) **The propagator as bracket.** The propagator S^{ij} of special geometry is the homotopy transfer data for the Hodge-to-de-Rham spectral sequence on $PV^{\bullet,\bullet}$. Concretely:

$$[\Theta^{(r)}, \Theta^{(g-r)}]_i = \bar{C}_i^{jk} D_j \Theta^{(r)} \cdot D_k \Theta^{(g-r)},$$

so the Lie bracket in $\mathcal{L}_X$ is “propagator-dressed.”

- (v) **The non-holomorphic completion.** The fact that $\Theta^{(g)}$ is not holomorphic (the anomaly) is the statement that Θ is a MC element of the full dgLa $\mathcal{L}_X$, not of the holomorphic sub-dgLa $PV^{\bullet,0}$. Holomorphic ambiguity (the freedom in completing F_g to a modular object) corresponds to the freedom in choosing a representative within the MC gauge-equivalence class.

Proposition 3.3 (MC = BCOV, schematic). *The Maurer–Cartan equation $D\Theta + \frac{1}{2}[\Theta, \Theta] = 0$ in the dgLa $\mathcal{L}_X = PV^{\bullet,\bullet}(X^\vee)[[\hbar]]$, expanded at order $\hbar^{2g-2}$, reproduces the BCOV holomorphic anomaly equation (??). The propagator S^{ij} of special geometry is the contraction kernel of the Lie bracket; the Yukawa coupling C_{ijk} is the structure constant of the Schouten–Nijenhuis bracket; and the Kähler covariant derivative D_i is the dgLa differential restricted to the holomorphic tangent directions.*

Remark 3.4. This is, of course, the perspective championed by Costello–Li, who construct the BCOV theory as a quantization of the Kodaira–Spencer dgLa. What we add is the identification of this dgLa with the chiral bar dgLa of Volume I: the BCOV complex $\mathcal{L}_X$ is quasi-isomorphic to the dgLa $C^\bullet(B(A_X))$ governing deformations of the bar complex of the quantum vertex chiral group $G(X)$. In this identification, the BCOV MC element Θ is the shadow obstruction tower Θ_{A_X} .

3.4. The holomorphic anomaly and the shadow obstruction tower. The holomorphic anomaly has a clean interpretation in the shadow obstruction tower:

B-model	Shadow obstruction tower (Vol I)
Holomorphic limit ($\bar{t} \rightarrow \infty$)	Tree-level (genus 0) shadow
Anti-holomorphic dependence of F_g	Genus- g MC obstruction
Propagator S^{ij}	Homotopy transfer / degeneration data
Yukawa C_{ijk}	Cubic shadow C (arity 3)
Holomorphic ambiguity	MC gauge freedom
Modular completion	Full automorphic correction

The “holomorphic limit” of the B-model (sending $\bar{t} \rightarrow \infty$, i.e. restricting to the large-complex-structure point) corresponds to retaining only the tree-level shadow: the genus-0 data, which is purely holomorphic. Each genus $g \geq 1$ introduces anti-holomorphic dependence, which in the shadow obstruction tower corresponds to the g -th obstruction class living in a cohomology group that is not concentrated in the holomorphic sector.

4. THE TOPOLOGICAL VERTEX FOR TORIC CY3

4.1. Toric CY3 and their diagrams. A toric CY3 X_Σ is determined by a fan Σ in $\mathbb{Z}^3$ with all generators coplanar (the CY condition). The toric diagram is the projection of the fan to a convex polygon in $\mathbb{Z}^2$. The geometry of X_Σ is encoded in the trivalent planar graph dual to the toric diagram: vertices correspond to $\mathbb{C}^3$ patches, edges correspond to $\mathbb{C}^2$ patches (“gluing strips”), and the external legs correspond to non-compact divisors.

Example 4.1 ($\mathbb{C}^3$: the simplest vertex). A single vertex. $X = \mathbb{C}^3$, quiver = Jordan quiver, $G(X) \supset Y(\widehat{\mathfrak{gl}}_1) \simeq \mathcal{W}_{1+\infty}$.

Example 4.2 (Resolved conifold). Two vertices joined by an internal edge. $X = \mathcal{O}(-1) \oplus \mathcal{O}(-1) \rightarrow \mathbb{P}^1$. The partition function $Z = M(q)^2 \sum_\lambda q^{|\lambda|} Q^{|\lambda|} (s_\lambda(q^\rho))^2$ where $M(q) = \prod_{n=1}^\infty (1 - q^n)^{-n}$ is the MacMahon function.

4.2. The AKMV topological vertex. The fundamental result of Aganagic–Klemm–Mariño–Vafa (AKMV) is that the A-model topological string partition function on any toric CY3 X factorizes over the vertices and edges of the toric diagram:

$$(8) \quad Z(X) = \sum_{\{\lambda_e\}} \prod_{\text{vertices } v} C_{\lambda_{e_1} \lambda_{e_2} \lambda_{e_3}}(q) \prod_{\text{edges } e} (-Q_e)^{|\lambda_e|} f_{\lambda_e}(q),$$

where:

- The sum is over assignments of partitions λ_e to each internal edge e of the toric diagram;
- $C_{\lambda\mu\nu}(q)$ is the *topological vertex* — a universal amplitude attached to each trivalent vertex, depending on the three partitions assigned to its edges;
- $Q_e = e^{-t_e}$ is the Kähler parameter of the edge e (the exponentiated area of the corresponding $\mathbb{P}^1$);
- $f_\lambda(q) = (-1)^{|\lambda|} q^{(\kappa_\lambda)/2}$ is the framing factor, with $\kappa_\lambda = \sum_i \lambda_i(\lambda_i - 2i + 1)$.

The topological vertex itself has the explicit expression (in the Schur-function basis):

$$(9) \quad C_{\lambda\mu\nu}(q) = q^{\frac{\kappa_\mu}{2}} s_{\nu^t}(q^\rho) \sum_{\eta} s_{\lambda^t/\eta}(q^{\nu+\rho}) s_{\mu/\eta}(q^{\nu^t+\rho}),$$

where $q^\rho = (q^{-1/2}, q^{-3/2}, q^{-5/2}, \dots)$ and $q^{\nu+\rho} = (q^{\nu_1-1/2}, q^{\nu_2-3/2}, \dots)$ are specializations of symmetric functions, and $s_{\lambda/\eta}$ are skew Schur functions.

4.3. The topological vertex as local root datum. The topological vertex $C_{\lambda\mu\nu}(q)$ is a “local root datum” of $G(X)$. The precise formulation:

Construction 4.3 (Local root datum). To each trivalent vertex v of the toric diagram, associate the local quantum vertex chiral group $G_v = G(\mathbb{C}^3)$, with:

- (i) **Lattice:** $\Lambda_v = K_0(\text{pt}) = \mathbb{Z}$ (the dimension vector).

- (ii) **Root datum:** the affine Yangian $Y(\widehat{\mathfrak{gl}}_1)$ root system. The root multiplicities $\text{mult}_v(n) = p(n)$ (partitions of n), controlled by the MacMahon function $M(q)$.
- (iii) **Vertex amplitude:** the topological vertex $C_{\lambda\mu\nu}(q)$ is the matrix element of the intertwiner

$$\Phi_v: \mathcal{F}_\lambda \otimes \mathcal{F}_\mu \otimes \mathcal{F}_\nu \rightarrow \mathbb{C}$$

in the Fock-space representation $\mathcal{F}_\lambda$ of G_v , where $\mathcal{F}_\lambda$ is the Fock module labeled by the partition λ .

- (iv) **The shadow:** the $\mathbb{C}^3$ partition function $Z(\mathbb{C}^3) = C_{\emptyset\emptyset\emptyset}(q) = M(q) = \prod_{n=1}^{\infty} (1-q^n)^{-n}$ is the denominator identity of G_v .

Remark 4.4 (MacMahon and DT). The MacMahon function $M(q) = \sum_{n=0}^{\infty} p_3(n) q^n$, which counts 3D partitions, is the DT partition function of $\mathbb{C}^3$ (by the DT/GW correspondence of Maulik–Nekrasov–Okounkov–Pandharipande). Its role as the denominator identity of the local root datum of $G(\mathbb{C}^3)$ is the simplest instance of Theorem CY-B.

4.4. Gluing = tensor product of root data. The factorization (??) has a clean interpretation in terms of root data:

Construction 4.5 (Gluing of root data). The global root datum $\mathcal{R}(X)$ is assembled from local root data $\mathcal{R}_v$ by the following procedure.

- (i) **At each vertex** v : the local root datum $\mathcal{R}_v = \mathcal{R}(\mathbb{C}^3)$ with its vertex amplitude $C_{\lambda\mu\nu}$.
- (ii) **At each edge** e **joining vertices** v, w : the “propagator” is the edge weight $(-Q_e)^{|\lambda_e|} f_{\lambda_e}(q)$, which implements the contraction

$$\mathcal{F}_{\lambda_e}^{(v)} \otimes (\mathcal{F}_{\lambda_e}^{(w)})^* \xrightarrow{\text{contract}} \mathbb{C},$$

i.e., the identification (tensor product pairing) of the Fock-space representation of G_v with the dual of the Fock-space representation of G_w along the shared edge.

- (iii) **The lattice:** $\Lambda(X) = \bigoplus_e \mathbb{Z} \cdot [e]$ (one generator per internal edge = one Kähler class), plus the embedding $\Lambda_v \hookrightarrow \Lambda(X)$ at each vertex.
- (iv) **The global quantum group:** $G(X)$ is the tensor product (over the toric diagram) of the local G_v ’s, quotiented by the edge identifications. In the language of factorization algebras, this is the factorization product over the toric diagram.

The summation over partitions λ_e in (??) is the “path integral” over the internal degrees of freedom: it sums over all representations propagating along each edge, weighted by the Kähler factor $Q_e^{|\lambda_e|}$. This is the trace over the tensor product:

$$(10) \quad Z(X) = \text{Tr}_{\bigotimes_e \mathcal{F}_e} \left(\prod_v \Phi_v \cdot \prod_e (-Q_e)^{L_e^{(0)}} q^{L_e} \right),$$

where $L_e^{(0)}$ and L_e are the zero-mode and total energy operators on the edge Fock spaces.

Remark 4.6 (Relation to the crystal melting picture). The factorization over vertices is the algebraic incarnation of the crystal melting picture of Okounkov–Reshetikhin–Vafa: 3D partitions label states in the Fock space, the vertex operator creates/annihilates boxes, and the partition function counts weighted crystal configurations. The root datum language merely repackages this as data of the quantum vertex chiral group.

Conjecture 4.7 (Topological vertex as chiral intertwiner). *The topological vertex $C_{\lambda\mu\nu}(q)$ coincides with the matrix element of the E_2 -chiral intertwiner of $G(\mathbb{C}^3) \simeq Y(\widehat{\mathfrak{gl}}_1) \simeq \mathcal{W}_{1+\infty}$:*

$$C_{\lambda\mu\nu}(q) = \langle \lambda | \Phi_{A_{\mathbb{C}^3}}^{E_2}(z_1, z_2, z_3) | \mu, \nu \rangle,$$

where $\Phi_{A_{\mathbb{C}^3}}^{E_2}$ is the genus-0, 3-point amplitude of the E_2 -chiral algebra $A_{\mathbb{C}^3} = \Phi(\mathcal{C}_{\mathbb{C}^3})$, and the Fock states $|\lambda\rangle$ are identified with basis elements of the bar complex $B(A_{\mathbb{C}^3})$.

5. CONNECTION TO THE VOLUME I SHADOW OBSTRUCTION TOWER

5.1. The leading term: $F_g = \kappa \cdot \lambda_g$. Volume I establishes that for any chiral algebra A with modular characteristic $\kappa(A)$, the genus- g obstruction on the *uniform-weight lane* (the component of highest Hodge weight in $H^\bullet(\overline{\mathcal{M}}_g)$) is:

$$\text{obs}_g(A)|_{\text{uniform}} = \kappa(A) \cdot \lambda_g,$$

where λ_g is the top Chern class of the Hodge bundle $\mathbb{E}_g \rightarrow \overline{\mathcal{M}}_g$. For the quantum vertex chiral group $G(X)$ of a CY3, the modular characteristic is conjecturally:

$$\kappa(G(X)) = \chi^{\text{CY}}(\mathcal{C}),$$

the categorical CY Euler characteristic (this identification requires $G(X)$ to exist, which is Conjecture CY-A₃). For compact CICYs with $h^{1,0} = 0$, this coincides with $\chi_{\text{top}}(X)/24$, but in general κ depends on the full chiral algebra and can differ drastically from any simple function of χ_{top} (see the modular trace chapter and Proposition ??; e.g. $\kappa(K3 \times E) = 5$ while $\chi_{\text{top}} = 0$).

This gives the leading term of the genus- g topological string amplitude:

$$(11) \quad F_g|_{\text{leading}} = \kappa(G(X)) \cdot \int_{\overline{\mathcal{M}}_g} \lambda_g \cdot (\text{known intersection number}).$$

For $g \geq 2$, the Faber–Pandharipande formula gives:

$$\int_{\overline{\mathcal{M}}_{g,1}} \psi_1^{2g-2} \lambda_g = \frac{2^{2g-1} - 1}{2^{2g-1}} \cdot \frac{|B_{2g}|}{(2g)!},$$

where B_{2g} is the Bernoulli number. The constant-map contribution to F_g (the $\beta = 0$ term) is indeed:

$$(12) \quad F_g^{\beta=0} = \frac{(-1)^{g-1} \chi(X)}{2} \cdot \frac{|B_{2g}| |B_{2g-2}|}{2g(2g-2)(2g-2)!},$$

which is precisely $\kappa(G(X)) \cdot \lambda_g$ evaluated via the Mumford relation.

5.2. Shadow corrections beyond the leading term. The full genus- g amplitude $F_g = \sum_{\beta} N_{g,\beta} Q^\beta$ includes contributions from all curve classes $\beta \neq 0$. These are the *shadow corrections* to the leading term:

$$(13) \quad F_g = \underbrace{\kappa(G(X)) \cdot \lambda_g}_{\text{uniform-weight lane}} + \underbrace{\sum_{\beta \neq 0} N_{g,\beta} Q^\beta}_{\text{shadow corrections}}.$$

The shadow corrections are organized by the root lattice of $G(X)$:

- (i) The β -th Fourier coefficient $N_{g,\beta}$ is the genus- g amplitude of the root β in the shadow obstruction tower. For $g = 0$, these are the tree-level structure constants (Yukawa couplings). For $g = 1$, they include the genus-1 one-loop correction controlled by the Ray–Singer torsion.
- (ii) The Q^β fugacity is the root-lattice fugacity: $Q^\beta = e^{-\langle t, \beta \rangle}$ where t is the point in the Kähler moduli space (= the Cartan element of $G(X)$).
- (iii) The shadow obstruction tower $\Theta_{A_X}^{\leq r}$ truncated at arity r captures the F_g up to corrections from roots of “arity $> r$ ” (roughly, those β whose length in the root lattice exceeds r).

Remark 5.1 (The three towers). Three expansions of the topological string free energy are unified:

- (1) **Genus expansion:** $F = \sum_g F_g g_s^{2g-2}$. This is the shadow obstruction tower graded by genus.
- (2) **Instanton expansion:** $F_g = \sum_\beta N_{g,\beta} Q^\beta$. This is the root-lattice decomposition of the genus- g shadow.
- (3) **BPS expansion:** $F = \sum_{g,\beta,k} \frac{n_\beta^g}{k} (2 \sin \frac{k g_s}{2})^{2g-2} Q^{k\beta}$. This is the Gopakumar–Vafa reformulation, which in the root datum language gives the multiplicities n_β^g as the intrinsic BPS content of each root.

5.3. The partition function as denominator identity. For the total partition function $Z = e^F$, the product structure of the GV formula gives:

$$(14) \quad Z = M(q)^{\chi(X)/2} \cdot \prod_{\beta > 0} \prod_{k=1}^{\infty} \prod_{j=-\infty}^{\infty} \left(1 - q^{j+k/2} Q^{k\beta}\right)^{(-1)^{2j+1} n_\beta^{|j|}},$$

(schematically), which is a Weyl–Kac–Borcherds-type infinite product — the denominator identity of the quantum vertex chiral group $G(X)$.

- The leading factor $M(q)^{\chi(X)/2}$ is the MacMahon function raised to the power $\chi(X)/2$. This exponent coincides with $\kappa(G(X))$ only when $\kappa = \chi(X)/2$, which holds for toric CY3 (where the Heisenberg subalgebra dominates) but fails in general: for $K3 \times E$, $\chi = 0$ while $\kappa = 5$ (AP48). The $M(q)^{\chi(X)/2}$ factor is the constant-map (Weyl vector) contribution.
- The product over roots $\beta > 0$ with multiplicities n_β^g is the automorphic correction: the imaginary roots of $G(X)$ with their BPS multiplicities.
- For $K3 \times E$: Z is related to $(\Delta_5)^{-2}$, and the product formula is the Borcherds denominator identity of $\mathfrak{g}_{\Delta_5}$.
- For toric CY3: Z is the DT partition function, factorized via the topological vertex.

6. SUMMARY: THE COMPLETE DICTIONARY

We collect the full dictionary between the topological string on a CY3 X and the quantum vertex chiral group $G(X)$.

Topological string	$G(X)$ / Vol III	Vol I
String coupling g_s	Quantization parameter $\hbar$	$\hbar$
Kähler parameter Q^β	Root-lattice fugacity	—
Genus- g amplitude F_g	Genus- g obstruction	$\text{obs}_g(A_X)$
GW invariant $N_{g,\beta}$	Shadow amplitude at β	$\Theta^{(g)} _\beta$
GV invariant n_β^g	BPS root multiplicity	Intrinsic root content
BCOV propagator S^{ij}	Homotopy transfer kernel	Degeneration data
Yukawa C_{ijk}	Cubic shadow (arity 3)	C in $\Theta^{\leq 3}$
Hol. anomaly equation	MC at genus g	$D\Theta^{(g)} + \dots = 0$
Holomorphic ambiguity	MC gauge freedom	Shadow gauge
$F_g^{\beta=0}$ (constant maps)	$\kappa \cdot \lambda_g$	Uniform-weight lane
$F_g^{\beta \neq 0}$ (instantons)	Shadow corrections	Off-diagonal shadow
Partition function $Z = e^F$	Denominator identity	Bar-complex Euler product
Topological vertex $C_{\lambda\mu\nu}$	Local root datum	E_2 -chiral intertwiner
Edge gluing	Tensor product of root data	Factorization product
Mirror symmetry	HMS: $\text{Fuk}(X) \simeq D^b(\text{Coh}(X^\vee))$	Koszul duality
DT/GW correspondence	Cohomological Hall $\leftrightarrow$ GW	$E_1 \hookrightarrow E_2$

7. OPEN QUESTIONS AND FUTURE DIRECTIONS

- Q1. Full proof of Conjecture ??.** The identification of GW invariants with shadow amplitudes is established at the level of the leading term $(\kappa \cdot \lambda_g)$ and for toric CY3 (via the topological vertex). The general case requires constructing the virtual fundamental class from the shadow obstruction tower — this is the “quantization” step in the CY-to-chiral functor.
- Q2. Refined topological vertex and motivic root data.** The refined topological vertex $C_{\lambda\mu\nu}(q, t)$ (Iqbal–Kozcaz–Vafa), which computes the motivic DT invariants, should correspond to the motivic refinement of the root datum — replacing integer multiplicities by classes in the Grothendieck ring of varieties.
- Q3. Compact CY3 and the holomorphic anomaly.** For compact CY3, the BCOV equation is essential (unlike for toric CY3, where the B-model is a free theory and F_g is holomorphic for $g \geq 2$ in appropriate coordinates). Making Construction ?? fully rigorous requires identifying the Costello–Li BCOV theory with the chiral bar dgLa of $G(X)$.
- Q4. Non-perturbative completion.** The genus expansion $F = \sum_g F_g g_s^{2g-2}$ is asymptotic (the F_g grow as $(2g)!$). The non-perturbative completion should come from the resurgent structure of the shadow obstruction tower, with Borel singularities corresponding to the real roots of $G(X)$ (large-radius instantons).

- Q5. The $K3 \times E$ topological string.** The topological string on $K3 \times E$ is not toric, so the topological vertex does not directly apply. However, the BKM superalgebra $\mathfrak{g}_{\Delta_5}$ provides the root datum, and the Igusa cusp form Δ_5 is the denominator identity. Constructing F_g from the shadow obstruction tower in this case should recover the Oberdieck–Pixton formula (cf. the $K3 \times E$ chapter of Volume III).
- Q6. Relation to wall-crossing.** The DT/GW correspondence and the wall-crossing formula of Kontsevich–Soibelman should be visible in the root datum as changes of Weyl chamber — the shadow obstruction tower Θ_{A_X} should be wall-crossing invariant, with its chamber-dependent expansion recovering the different BPS spectra.

8. THE MACMAHON OBSTRUCTION

The MacMahon function $M(q) = \prod_{n \geq 1} (1 - q^n)^{-n}$ is the DT partition function of $\mathbb{C}^3$ and the denominator identity of the $\mathbb{C}^3$ quantum vertex chiral group. A natural question: does the shadow obstruction tower of $A_{\mathbb{C}^3}$ reproduce $M(q)$ term by term?

The swarm computation (`compute/lib/macmahon_shadow_decomposition.py`, 50 tests) gives a precise negative answer.

8.1. The mismatch. Write the genus expansion of both objects. The topological string free energy of $\mathbb{C}^3$ is $F = \sum_{g \geq 0} F_g g_s^{2g-2}$, and the shadow obstruction tower has projections $\Theta^{(g)} = \kappa \cdot \lambda_g + (\text{higher arity})$. At each genus g , the topological string coefficient is

$$F_g^{\text{top}}(\mathbb{C}^3) = \frac{(-1)^{g-1} B_{2g}}{2g(2g-2)!} \quad (g \geq 2),$$

involving a *single* Bernoulli number B_{2g} . The shadow obstruction tower, by contrast, produces

$$F_g^{\text{sh}} = \kappa \cdot \lambda_g^{FP} = \kappa \cdot \frac{(-1)^{g-1} B_{2(g-1)} \cdot B_{2g}}{(2g)! \cdot (2g-2)} \quad (g \geq 2),$$

involving the *product* $B_{2(g-1)} \cdot B_{2g}$ (the Faber–Pandharipande class). These differ: the shadow tower has a double Bernoulli product where the topological string has a single Bernoulli number.

8.2. Generating-function-level agreement. Despite the genus-by-genus mismatch, the two objects agree at the level of generating functions. Let

$$\hat{A}(x) = \frac{x/2}{\sinh(x/2)} = 1 - \frac{x^2}{24} + \frac{7x^4}{5760} - \dots$$

be the $\hat{A}$ -genus. Then:

- The shadow tower generating function is $\sum_{g \geq 1} F_g^{\text{sh}} \hbar^{2g} = \kappa \cdot (\hat{A}(i\hbar) - 1)$.
- The topological string generating function $\sum_{g \geq 2} F_g^{\text{top}} g_s^{2g-2}$ is the asymptotic expansion of $\log M(e^{-g_s})$.

The relation between the two is through the Laplace/Borel transform: the double Bernoulli product $B_{2(g-1)} B_{2g}$ is the convolution of two single-Bernoulli series. The $\hat{A}$ -genus packages this convolution into a closed form that is *not* the same as the MacMahon expansion genus by genus, but whose Borel sum agrees.

8.3. The obstruction.

Warning 8.1 (MacMahon $\neq$ scalar shadow, term by term). $M(q)$ is NOT the scalar shadow tower evaluated term by term. The scalar shadow $F_g^{\text{sh}} = \kappa \lambda_g^{FP}$ differs from $F_g^{\text{top}}(\mathbb{C}^3)$ at every genus $g \geq 2$. The agreement is at the level of generating functions (Borel-summed), not at the level of individual Taylor coefficients.

The physical interpretation: the shadow tower computes the *algebraic* genus expansion (from the bar complex on $\overline{\mathcal{M}}_{g,n}$), while the MacMahon function comes from the *analytic* partition function (the regularized Euler product on the root lattice). The passage from algebraic to analytic requires a resummation that reshuffles genus-by-genus coefficients.